

Multiple Choice (10 marks)

1. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan
2. Can we build a secure PRP from a secure PRG?
 - (a) No, it cannot be done
 - (b) Yes, just plug the GGM PRF into the Luby-Rackoff theorem
 - (c) It depends on the underlying PRG
 - (d) Option text
3. Which of the following are most vulnerable to injection attacks?
 - (a) Session IDs
 - (b) Registry keys
 - (c) Network communications
 - (d) SQL queries based on user input
4. Why apps developed in languages like C, C++ is prone to Buffer-overflow?
 - (a) No string boundary checks in predefined functions
 - (b) No storage check in the external memory
 - (c) No processing power check
 - (d) No database check
5. Which of the following is not a block cipher operating mode?
 - (a) ECB
 - (b) CFB
 - (c) CBF
 - (d) CBC

True / False (10 marks)

Answer True or False

6. True or False: A buffer is a sequential segment of memory allocated for storing data, such as character strings or arrays of integers.
7. True or False: Blackbox fuzzers can effectively cover every line of code in a program due to their ability to analyze source code.
8. True or False: Buffer-overflow issues can be effectively mitigated solely through comprehensive memory checks without boundary checks.
9. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol used for providing confidentiality and integrity in network communications.
10. True or False: Nmap identifies port states as active, inactive, and standby when assessing network security.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the classic security properties in cybersecurity and critically evaluate why availability is often considered distinct from the other fundamental properties.
12. Discuss the role of authorization in cybersecurity, focusing on how it restricts user access to operations and data, and analyze its significance in protecting sensitive information.

End of Paper